A Curvature Identity for the Fibonacci Digital-Root Cycle Modulo 9

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Abstract

For each seed we form the 24-periodic orbit of the Fibonacci recurrence modulo 9, write its terms with the digital-root convention $0 \mapsto 9$, and measure its *digital curvature* $S = \sum_n \kappa_n^2$, the sum of squared discrete second differences. On the 72 full-period seeds modulo 9 the curvature takes exactly three values, $\{648, 972, 1296\}$, each on 24 seeds; the maximum 1296 is attained precisely by the classical Fibonacci cycle $(1, 1)$ and its rotations. This is a finite statement and we prove it by exhaustive enumeration. The curvature is the squared norm of a circulant operator and therefore has an exact Fourier form, $S = \frac{16}{N} \sum_k \sin^4(\pi k/N)\, |\hat{d}_k|^2$, which makes it a high-pass energy. Modulo 9 the Fibonacci class is also the class with the largest share of its curvature in the near-Nyquist half of the spectrum (about 98%), but this describes the maximizer rather than explaining it: at other moduli neither the near-Nyquist energy nor its share reliably identifies the maximizing orbit. We also show that modulo 9 the rule $0 \mapsto 9$ does not affect S at all—raw residues $0, \ldots, 8$ give the same three values—because every orbit negates itself after half a period. Finally we report how far the maximality property extends to other moduli (it is a small-modulus phenomenon, not a law) and retire a conjecture from earlier versions of this note. This version corrects errors in Versions 1–4; see the revision history in Section 8.

Keywords: Fibonacci numbers; Pisano period; digital root; discrete curvature; circulant operator; discrete Fourier transform.

1 Introduction

Let $F(n)$ be the Fibonacci numbers, $F(0) = 0$, $F(1) = 1$. The sequence $F(n) \bmod 9$ has Pisano period 24 [1]. Replacing each residue 0 by 9 (the digital-root convention used in base 10, since the digital root of a positive integer equals its value modulo 9 with $0 \mapsto 9$; the digital root of 0 itself is 0 [2]) yields the familiar cycle

$$1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9.$$

This 24-tuple is well known, but its higher-order difference statistics are not usually examined. We attach to every modulo-9 orbit a single number—its *digital curvature*, the energy of its discrete second difference—and ask what values that number takes.

The answer modulo 9 is a clean trichotomy (Theorem 1). The curvature functional is the squared norm of the cyclic second-difference operator, which is circulant and hence diagonalized by the discrete Fourier transform. This gives an exact spectral formula for S (Proposition 2), which turns the opaque integer S into an interpretable high-pass energy. The formula does not by itself prove the classification, which rests on enumeration, and at general moduli it does not identify the maximizing orbit (Section 5); we use it as a description, not a derivation. The spectral identity is a standard fact of harmonic analysis on the cycle; what is (to our knowledge) not previously recorded is its application to the digital-root orbits of a Pisano period.

We have tried to be careful about what is and is not established. The trichotomy is a theorem proved by finite computation. The spectral identity and the half-period negation property (Proposition 1) are theorems. The statement that the Fibonacci orbit maximizes curvature is true for small moduli but fails for larger ones, and we report it as the bounded empirical fact that it is. Earlier versions of this note advanced a closed-form conjecture and a symmetry argument that turn out to be, respectively, a numerical coincidence and an error; Version 4 in turn overstated what the spectral picture explains and what had been verified. All of these are corrected below.

2 Definitions

Fix a modulus $m \geq 2$ (we take $m = 9$ unless stated otherwise) and let $P = \pi(m)$ be its Pisano period [1, 3]. For a seed $(a_0, a_1) \in (\mathbb{Z}/m\mathbb{Z})^2$ generate the orbit

$$a_{n+1} = a_n + a_{n-1} \pmod{m}.$$

We restrict attention to seeds whose orbit has *minimal period exactly* P ("full-period" seeds); these are the orbits that realize the full Pisano cycle. The *rotation class* of such a seed is the set of P pairs (a_n, a_{n+1}) occurring along its orbit—equivalently, its P cyclic shifts. Display each residue with

$$d(r) = \begin{cases} m & r \equiv 0, \\ r & r \in \{1, \ldots, m-1\}, \end{cases}$$

so displayed values lie in $\{1, \ldots, m\}$. For $m = 9$ this is the base-10 digital root; for other m it is simply a choice of integer representatives. Write $r_n \in \{0, \ldots, m-1\}$ for the raw residues, so $d_n = d(r_n)$, and set

$$\kappa_n = d_{n+1} - 2\,d_n + d_{n-1} \in \mathbb{Z}, \qquad S = \sum_{n=0}^{P-1} \kappa_n^2,$$

all indices taken cyclically modulo P. We call S the *digital curvature* of the orbit. Two remarks and one proposition are worth making at the outset.

Remark 1 (Choice of representatives). In general S depends on which integer representatives of the residues are used. For example, the orbit of $(1,1)$ modulo 11 has $S = 638$ with representatives in $\{0,\dots,10\}$ but $S = 1430$ with the display $\{1,\dots,11\}$. Modulo 9, however, these two conventions give the same S on every orbit (Proposition 1), while the symmetric representatives $\{-4,\dots,4\}$ give yet another value (648 for the $(1,1)$ orbit). So S is an invariant of a chosen integer presentation of the orbit, not of the abstract residue sequence. We fix the digital-root presentation throughout.

Proposition 1 (Half-period negation)**.** *Let* $M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$*, the matrix that sends each pair* $(a_n, a_{n+1})^{\mathsf{T}}$ *of an orbit to the next pair* $(a_{n+1}, a_{n+2})^{\mathsf{T}}$*. Suppose* P *is even and* $M^{P/2} \equiv -I \pmod{m}$*. Then on every orbit*

$$d_n = m - r_{n+P/2} \qquad \text{for all } n,$$

so $(Dd)_n = -(Dr)_{n+P/2}$ *and the displayed and raw orbits have the same curvature* S*. Their curvature spectra agree mode by mode:* $|\widehat{Dd}_k| = |\widehat{Dr}_k|$ *for every* k*. Modulo* 9 *the hypothesis holds, since* $M^{12} = \begin{pmatrix} F_{11} & F_{12} \\ F_{12} & F_{13} \end{pmatrix} = \begin{pmatrix} 89 & 144 \\ 144 & 233 \end{pmatrix} \equiv -I \pmod{9}$.

Proof. $M^{P/2} \equiv -I$ gives $r_{n+P/2} \equiv -r_n \pmod{m}$. If $r_n \neq 0$ then $r_{n+P/2} = m - r_n$, so $m - r_{n+P/2} = r_n = d_n$; if $r_n = 0$ then $r_{n+P/2} = 0$, so $m - r_{n+P/2} = m = d_n$. The second difference annihilates the constant m, so $(Dd)_n = -(Dr)_{n+P/2}$. A sign and a cyclic shift change neither the sum of squares nor the modulus of any Fourier coefficient. □

Among the 118 moduli $m \leq 120$ with even Pisano period, $M^{P/2} \equiv -I$ holds for 43, and the $(1,1)$ orbit has equal displayed and raw curvature at exactly those 43. For every $m \leq 300$ with even Pisano period we found no case of equal curvature without $M^{P/2} \equiv -I$, but we have not proved the converse.

Remark 2 (Curvature is a cyclic energy). $\kappa_n = (Dd)_n$ where D is the cyclic second-difference operator, so $S = \|Dd\|^2$ is the discrete Dirichlet-type energy of the displayed orbit on the cycle C_P. This observation drives everything in Section 4.

3 The Trichotomy Modulo 9

Theorem 1. *Among the* 72 *full-period seeds modulo* 9*, the digital curvature takes exactly the three values*

$$S \in \{648,\ 972,\ 1296\},$$

each attained by exactly 24 *seeds (equivalently, by one full-period orbit apiece). Moreover* $S = 1296$ *if and only if the seed lies in the rotation class of* $(1,1)$*, and then*

$$S = 1296 = \sum_{n=0}^{23} \kappa_n^2.$$

Proof. There are 81 ordered seeds $(a_0, a_1) \in \{1, \ldots, 9\}^2$; of these, 72 generate orbits of minimal period 24 and 9 do not. Evaluating S on each full-period orbit is a finite computation, carried out in the script of Section 7; it returns the multiset {648:24, 972:24, 1296:24}, with the value 1296 occurring exactly on the 24 rotations of the $(1, 1)$ orbit. A finite exhaustive check is a complete proof of a statement quantified over a finite set. □

The split of the 81 seeds into 72 of period 24, 8 of period 8 (both entries divisible by 3) and the fixed seed $(9, 9)$ is already recorded in the comments to OEIS A030132 [2]; what is new here is the curvature. By Proposition 1, the same three values arise if the orbits are written with residues in $\{0, \ldots, 8\}$ instead.

We stress that the enumeration *is* the proof. Earlier versions of this note tried to derive the three values from a congruence argument; that argument is vacuous (all three values are divisible by 9, so reduction modulo 9 cannot distinguish them) and has been removed. What survives from it is a correct structural observation about *which* orbits carry the three values.

Remark 3 (The three values are the three unit classes). The unit group $(\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 5, 7, 8\}$ acts on orbits by $a_n \mapsto g\, a_n$. *At the level of residues*, $F_{n+12} \equiv -F_n \pmod 9$, so negation $g = -1 \equiv 8$ maps the $(1, 1)$ orbit to its own shift by 12; hence the subgroup $\{\pm 1\}$ fixes each orbit and the six units collapse to the three classes

$$\{\pm 1\} = \{1, 8\}, \quad \{\pm 2\} = \{2, 7\}, \quad \{\pm 4\} = \{4, 5\},$$

i.e. $(\mathbb{Z}/9\mathbb{Z})^\times/\{\pm 1\}$. Scaling the $(1, 1)$ orbit by a representative of each class gives precisely the three curvature values:

$$g \in \{1, 8\} \to 1296, \qquad g \in \{2, 7\} \to 972, \qquad g \in \{4, 5\} \to 648,$$

verified directly. So the trichotomy is the statement that curvature is an invariant of $(\mathbb{Z}/9\mathbb{Z})^\times/\{\pm 1\}$, taking distinct values on its three classes. (This residue-level symmetry is what makes the unit-class action well defined. It does not survive as an exact antisymmetry of the second differences once the residues are written as integers in $\{1, \ldots, 9\}$, or in $\{0, \ldots, 8\}$; see Section 4.1.)

Table 1 records the second differences of the maximal $(1, 1)$ orbit, for use below. One checks directly that $\sum_n \kappa_n = 0$ and $\sum_n \kappa_n^2 = 1296$.

n	0	1	2	3	4	5	6	7	8	9	10	11
a_n	1	1	2	3	5	8	4	3	7	1	8	9
κ_n	8	1	0	1	1	−7	3	5	−10	13	−6	−2

n	12	13	14	15	16	17	18	19	20	21	22	23
a_n	8	8	7	6	4	1	5	6	2	8	1	9
κ_n	1	−1	0	−1	−1	7	−3	−5	10	−13	15	−16

Table 1: Second differences for the $(1, 1)$ orbit. $\sum \kappa_n = 0$, $\sum \kappa_n^2 = 1296$.

4 Curvature as Spectral Energy

The cyclic second-difference operator D on length-N sequences is circulant, hence diagonalized by the discrete Fourier transform [4]. This yields an exact formula for S.

Proposition 2. *Let $d = (d_0, \dots, d_{N-1})$ be the displayed orbit (of length $N = P$), with unnormalized DFT $\hat{d}_k = \sum_n d_n e^{-2\pi i k n/N}$. Then*

$$S = \frac{1}{N} \sum_{k=0}^{N-1} \mu_k^2 \, |\hat{d}_k|^2, \qquad \mu_k = 2\cos\frac{2\pi k}{N} - 2 = -4\sin^2\frac{\pi k}{N},$$

equivalently

$$\boxed{S = \frac{16}{N} \sum_{k=0}^{N-1} \sin^4\Big(\frac{\pi k}{N}\Big) \, |\hat{d}_k|^2.}$$

Proof. $\kappa_n = (Dd)_n = d_{n+1} - 2d_n + d_{n-1}$ with indices modulo N, so D is the circulant matrix with symbol $e^{2\pi i k/N} - 2 + e^{-2\pi i k/N} = 2\cos(2\pi k/N) - 2 = \mu_k$. The Fourier modes $e^{2\pi i k n/N}$ are its eigenvectors with eigenvalues μ_k, so $\widehat{Dd}_k = \mu_k \hat{d}_k$. By Parseval, $S = \|Dd\|^2 = \frac{1}{N}\sum_k |\mu_k \hat{d}_k|^2 = \frac{1}{N}\sum_k \mu_k^2 |\hat{d}_k|^2$, and $\mu_k^2 = 16\sin^4(\pi k/N)$. □

This is standard harmonic analysis on the cycle; we record it only because it converts the opaque integer S into something interpretable. The weight $16\sin^4(\pi k/N)$ vanishes at the constant mode $k = 0$, grows monotonically across $k = 1, \dots, \lfloor N/2 \rfloor$, and is maximal at the Nyquist mode $k = N/2$ (which exists because $N = \pi(m)$ is even for $m \geq 3$). Thus *digital curvature is a high-pass energy*: it ignores the orbit's mean and slow variation and rewards rapid, near-alternating oscillation. Call the modes with $\sin^2(\pi k/N) \geq \frac{1}{2}$—equivalently $N/4 \leq k \leq 3N/4$—the *near-Nyquist half* of the spectrum, and write

$$B = \frac{16}{N} \sum_{N/4 \leq k \leq 3N/4} \sin^4\Big(\frac{\pi k}{N}\Big) \, |\hat{d}_k|^2$$

for the part of S it carries. We call B the *near-Nyquist energy* and B/S the *near-Nyquist share*; the two can rank orbits differently (Section 5).

For the maximal $(1, 1)$ orbit ($N = 24$, $S = 1296$), Table 2 gives the per-mode contribution $c_k = \frac{1}{N}\mu_k^2 |\hat{d}_k|^2$. Conjugate modes k and $N-k$ contribute equally, so $S = c_0 + 2\sum_{k=1}^{11} c_k + c_{12}$. The near-Nyquist half carries about 97.6% of S for this orbit; the constant and low-frequency modes carry the remaining 2.4%.

Remark 4 (The maximizer is the highest-frequency class). The three modulo-9 classes are ordered by how much of their curvature sits in the near-Nyquist half: it is 97.6% of S for the 1296 class, against 94.0% and 86.6% for the 972 and 648 classes. So among the three full-period orbit classes modulo 9, the Fibonacci class is the one whose curvature is most concentrated near the Nyquist frequency. This is a diagnosis, not a derivation: because S *is* the high-pass energy, the statement reformulates maximality spectrally but does not explain why the $(1, 1)$ seed produces the most near-Nyquist spectrum in the first place (see Section 9). At other moduli the description does not even identify the maximizer (Section 5).

k	$16\sin^4(\pi k/24)$	$\lvert\hat{d}_k\rvert^2/N$	c_k
0	0.0000	570.375	0.000
1	0.0046	17.919	0.083
2	0.0718	3.375	0.242
3	0.3431	19.125	6.563
4	1.0000	3.375	3.375
5	2.1974	2.331	5.122
6	4.0000	3.375	13.500
7	6.3385	2.331	14.774
8	9.0000	3.375	30.375
9	11.6569	19.125	222.937
10	13.9282	3.375	47.008
11	15.4595	17.919	277.022
12	16.0000	3.375	54.000

Table 2: Spectral decomposition of the $(1,1)$ orbit's curvature. Conjugate modes k and $24-k$ contribute equally, so $S = c_0 + 2\sum_{k=1}^{11} c_k + c_{12} = 1296$. The weight $16\sin^4(\pi k/24)$ rises with frequency; the contributions c_k do not (they drop from $k=3$ to 4, 9 to 10, and 11 to 12). The dominant pairs are $k = 11, 13$ (277.0 each) and $k = 9, 15$ (222.9 each), with the Nyquist mode $k = 12$ contributing 54.0. The near-Nyquist half ($\sin^2(\pi k/24) \geq \frac{1}{2}$, i.e. $6 \leq k \leq 18$) accounts for about 97.6% of S.

4.1 The half-period asymmetry, corrected

The identity $F_{n+12} \equiv -F_n \pmod 9$, a case of Proposition 1, suggests the second differences should satisfy $\kappa_{n+12} = -\kappa_n$, which would force the two half-period curvature sums to be equal. *This is false*, and Version 2 of this note relied on it in error. The relation $\kappa_{n+12} = -\kappa_n$ holds for $n = 1, \ldots, 9$ but breaks at $n = 0, 10, 11$—exactly the index pairs adjacent to the zeros of the orbit at positions 11 and 23. Consequently the halves are *unequal*:

$$\sum_{n=0}^{11} \kappa_n^2 = 459, \qquad \sum_{n=12}^{23} \kappa_n^2 = 837, \qquad 459 + 837 = 1296,$$

and the entire imbalance $837 - 459 = 378$ comes from those three index pairs.

Version 4 attributed the break to the display rule $d(0) = 9$. That is not the cause. The residues do satisfy $r_{n+12} \equiv -r_n$, but integer representatives turn this into an exact antisymmetry only if the representative system is closed under negation. Neither $\{1, \ldots, 9\}$ nor $\{0, \ldots, 8\}$ is: with either one, the representatives of x and $-x$ sum to 9 when $x \neq 0$ but to 18 (respectively 0) when $x = 0$. Away from the zeros the constant offset 9 is invisible to D; next to them it is not. Raw residues therefore break the antisymmetry at the same three indices, with the two halves exchanged (837 and 459, as Proposition 1 predicts). With the symmetric representatives $\{-4, \ldots, 4\}$, which are closed under negation, the antisymmetry is exact and the halves are equal, 324 and 324, for a total of 648. In the digital-root presentation the curvature does not split evenly across the period, and the factorization $1296 = 2 \cdot 648$ is not explained by half-period symmetry.

5 How Far Maximality Extends

It is natural to ask whether the Fibonacci seed maximizes digital curvature for every modulus. It does not. Define, for any modulus m with even Pisano period, $S_{\max}(m)$ as the maximum of S over full-period orbits, computed with the same display convention.

Small moduli. Among moduli $m \leq 14$ with even Pisano period, $(1, 1)$ attains $S_{\max}(m)$ in every case; the first failure is $m = 15$.

Erosion with scale. Over the 118 moduli $m \leq 120$ with even Pisano period, $(1, 1)$ attains $S_{\max}(m)$ for 78 of them and fails for 40. The first three failures are $m = 15, 33, 40$ (and these are the only failures below 41); the full failure list is

$$15, 33, 40, 45, 49, 52, 53, 56, 59, 60, 64, 65, 67, 68, 71, 74, 80, 81, 83, 85,$$
$$86, 88, 90, 91, 92, 93, 97, 98, 103, 104, 105, 106, 107, 108, 109, 111, 115, 117, 118, 120.$$

The proportion of failures broadly increases with m (reaching 60–70% among moduli in the 80s–100s), though not monotonically.

No arithmetic predictor. No simple arithmetic rule we tested separates the failures from the successes. In particular "$5 \mid m$" fails badly in both directions (33 is a failure with $5 \nmid 33$; $20, 25, 30, 35$ are multiples of 5 for which $(1, 1)$ still wins).

What the spectral picture does not say. Version 4 claimed that "in every exceptional case we examined" the actual maximizer is the orbit with the most near-Nyquist energy. That is false under either reading of "most" (Section 4 defines the near-Nyquist energy B and share B/S).

- *Share.* At $m = 40$ the maximizer, the orbit of $(0, 3)$ with $S = 58320$, has share 93.7%, against 88.8% for $(1, 1)$ with $S = 51920$. But the orbit of $(0, 13)$ has the same curvature as $(1, 1)$, $S = 51920$, and share 96.1%. Among the 40 failures listed above, no curvature maximizer has the largest share in 24 cases, and at $m = 118$ the maximizer's share is slightly below that of $(1, 1)$ (93.94% against 93.99%).
- *Energy.* At $m = 71$ the maximizer, the orbit of $(0, 17)$, has $S = 242110$ and $B \approx 229974.2$, while the orbit of $(0, 29)$ has the smaller $S = 240406$ but the larger $B \approx 232216.4$. Over all 118 moduli $m \leq 120$ with even Pisano period, the curvature maximizer fails to maximize B at $m = 70, 71, 75, 91, 102$.

What does hold over this range is weaker: at each of the 40 failures the maximizer has strictly larger B than the $(1, 1)$ orbit. The spectral identity describes where the curvature of an orbit lives; it does not identify which orbit has the most.

6 What Does Not Hold

We record two negative results, both retiring claims from earlier versions, because closing off false directions is part of the contribution.

The closed form $S_{\max} = m(\pi(m)/2)^2$ is a coincidence. Among the moduli $m \leq 300$ with even Pisano period it holds at exactly $m = 9$ (1296) and $m = 27$ (34992); it also fails at $m = 729$. It fails already at $m = 3$ (actual 36, predicted 48), and on the next powers of three at $m = 81$ (actual 857304, predicted 944784) and $m = 243$ (actual 21677544, predicted 25509168). We have not proved that it fails for every larger m, but nothing in the data supports it. The appealing reading $1296 = 9F_{12} = 9 \cdot 144$ is numerology specific to $m = 9$. There is no closed form for $S_{\max}(m)$ supported by the data.

The phenomenon does not carry over to $a_{n+1} = a_n + 2a_{n-1}$. This recurrence is invertible modulo m exactly when m is odd. With the same display convention, for every odd m with $5 \leq m \leq 101$ the orbit of $(0, 1)$ (which contains $(1, 1)$) does not attain the largest S among orbits of the same minimal period; at $m = 3$ it is the only orbit of that period. Version 4 also made claims about the recurrence $a_{n+1} = 2a_n + a_{n-1}$ and the Tribonacci recurrence without stating definitions or a search range; we withdraw them.

7 Computational Verification

The script `papers/v5/reproduce.py` in the project repository asserts each exact numerical claim in this note, and each finite-range claim over the range stated: the trichotomy and seed-period split (Theorem 1); the representative comparisons of Remark 1 and Section 4.1; the half-period negation property and its finite-range companion (Proposition 1); the unit-class scalings (Remark 3); the second differences and half sums (Table 1); the spectral identity on five orbits (Proposition 2); the spectral table and the three near-Nyquist shares; the maximality sweep, full failure list and decade proportions; the near-Nyquist energy and share counterexamples and the count of failures without a unit-multiple maximizer (Sections 5 and 9); the closed-form check; and the $a_{n+1} = a_n + 2a_{n-1}$ sweep (Section 6). Spectral quantities are computed in floating point, and every comparison the script asserts has a margin far larger than rounding error. It uses only the Python standard library.

The original `verify_curvature.py`, which accompanies Versions 1–3, independently checks the trichotomy only. The Version 5 corrections follow an independent fact-check of Version 4, which re-derived its numerical claims with separate code.

8 Revision History and Errata

This is Version 5. It supersedes Versions 1–4 (doi:10.5281/zenodo.15811767), which remain available as the historical record.

Corrections in Version 5 (to Version 4).

- **The spectral maximizer claim was false.** Version 4 said that in every exceptional case it examined the curvature maximizer is the orbit with the most near-Nyquist energy, and asked for a proof that this always holds. Counterexamples exist for both the energy and the share readings, including $m = 40$ and $m = 71$ (Section 5).

- **The convention-dependence remark was false for $m = 9$.** Version 4 said that residues in $\{0, \ldots, m-1\}$ give a different S. Modulo 9 they give the same S on every orbit (Proposition 1); the dependence is real at other moduli (Remark 1).

- **The half-period asymmetry was misattributed.** It is caused by using representatives not closed under negation, not by the rule $d(0) = 9$; raw residues break it at the same indices (Section 4.1).

- **The verification claim was overstated.** Version 4 said its script reproduces every numerical claim and that every claim had been checked by two independent implementations. Its script did not test the spectral maximizer claim, the other-recurrence claims, or the unit-multiple count, printed the near-Nyquist percentages without asserting them, and checked the closed form only at five moduli; the original script checks only the trichotomy. Section 7 now states what is checked.

- **Finite searches were stated as universal.** "It never holds off the powers of three" is replaced by the range actually searched (Section 6).

- **The other-recurrence claims are made precise or withdrawn** (Section 6).

- **Wording.** The abstract and introduction said the spectral identity explains the classification and that the maximizer follows from it; it only reformulates it. The spectral-table caption said the contributions rise with frequency; only the weight does. The abstract wrote $\hat{a}_k$ for $\hat{d}_k$. The digital-root remark now says "positive integer" (the digital root of 0 is 0). The assumption $m \geq 2$ is stated, and the Nyquist mode is noted to exist only because $\pi(m)$ is even for $m \geq 3$. The failure list is now given in full. The seed-period split is credited to OEIS A030132.

Corrections in Version 4 (to Versions 1–3).

- **The trichotomy is proved by enumeration.** The congruence/"scaling" argument offered as a proof in Versions 1–3 is vacuous (all three curvature values are divisible by 9) and has been replaced by the finite verification, which is the actual proof. The valid residue of that argument—the three values as the three classes of $(\mathbb{Z}/9\mathbb{Z})^\times/\{\pm 1\}$—is kept as Remark 3.

- **The half-period symmetry claim was wrong.** Version 2 asserted $S = 2\sum_{n=0}^{11} \kappa_n^2$ from an antisymmetry that fails for these integer representatives; the true halves are 459 and 837 (Section 4.1). Version 3 already withdrew the explicit formula.

- **The conjecture $S_{\max} = m(\pi(m)/2)^2$ is false** (Section 6); it is a two-point coincidence at $m = 9, 27$.

- **Section 6's "explanation" was replaced** by the spectral identity (Section 4). Version 4 called the identity the reason the maximum is what it is; that overstated it (see above).

- **The novelty claim is softened.** We claim only that we are not aware of prior work applying this spectral/Dirichlet-energy lens to the digital-root orbits of a Pisano period; the underlying circulant diagonalization is classical. A search of OEIS (A030132 and neighbours) and of MathSciNet for "digital root" and "Pisano period" turned up no prior treatment of these second-difference statistics, but we make no exhaustive-priority claim.

9 Open Questions

- Characterize the moduli for which $(1, 1)$ fails to maximize curvature (Section 5). No arithmetic predictor was found, and 15 of the 40 failures with $m \leq 120$ have no maximizing orbit that is a unit multiple of the Fibonacci orbit.
- Find a criterion, spectral or arithmetic, that does identify the curvature maximizer; neither the near-Nyquist energy B nor the share B/S does (Section 5).
- Prove or refute the converse of Proposition 1: is the curvature of the $(1, 1)$ orbit the same in both presentations only when $M^{P/2} \equiv -I \pmod{m}$? This holds for every $m \leq 300$ with even Pisano period.
- Is the $S_{\max} = m(\pi(m)/2)^2$ coincidence at $m = 9, 27$ part of any larger pattern, or genuinely isolated?

Disclosure statement

The author reports no competing interests.

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